Project Journal — Abusive Language Detection in Social Networks

Jaime Morales

Project Progress Overview

This journal documents the development of an abusive language detection system across three milestones, from a single-domain proof-of-concept to a full cross-domain evaluation framework.

Milestone 1 established the research topic, the datasets to be used (Jigsaw and OLID), and the central research question: whether a model trained on formal Wikipedia-style text can generalise to informal Twitter-style text.

Milestone 2 implemented a three-script pipeline and fine-tuned RoBERTa-base on a balanced subset of 3,000 Jigsaw samples running on CPU. Validation F1 reached 0.9420 and test F1 reached 0.9082 on the in-domain Jigsaw test set. This result served as the baseline for the domain transfer experiments in Milestone 3.

Milestone 3 expands the system to the full Jigsaw dataset (159,571 samples) and adds OLID (11,916 tweets from Twitter). Three RoBERTa-base models are trained and evaluated in a 3×3 cross-domain design on Google Colab T4 GPU. The main finding is a 14-point F1 degradation when a Jigsaw-trained model is evaluated on Twitter data (0.8352 → 0.6932), confirming the domain shift hypothesis.

1. Software Architecture

The system is implemented as a flat, three-stage pipeline. Each stage is a standalone script that can be executed independently.

- **src/preprocess.py:** Loads the raw Jigsaw CSV, derives a binary label from the six toxicity columns, balances the dataset to 1,500 toxic and 1,500 clean samples, applies text cleaning (lowercasing, URL removal, special character removal), and splits the result into train (2,100), validation (300), and test (600) sets saved to data/processed/.
- **src/train.py:** Loads the processed splits, tokenizes the text using the RoBERTa tokenizer, fine-tunes RoBERTa-base for sequence classification using PyTorch, and saves the trained model weights to models/roberta-jigsaw/.
- **src/evaluate.py:** Loads the saved model, runs inference on the test set, and outputs accuracy, F1-score, a full classification report, and a confusion matrix saved to results/.

Data flows linearly: data/raw/ → data/processed/ → models/ → results/. No APIs, no abstraction layers, and no intermediate databases are used.

2. Technologies and Algorithms

The following technologies were selected for the implementation, each with a justification and at least one alternative.

Primary Model: RoBERTa-base

RoBERTa (Robustly Optimized BERT Pretraining Approach) is a transformer encoder pre-trained via Masked Language Modeling (MLM). It was chosen because its bidirectional attention captures context from both sides of a token, which is essential for detecting subtle or implicit abuse. The roberta-base variant (125M parameters) is manageable on CPU for small datasets.

Alternative: BERT-base. BERT is the predecessor of RoBERTa and shares the same transformer architecture. The main differences are that RoBERTa uses dynamic masking, removes the Next Sentence Prediction objective, and was trained on significantly more data (~160GB vs ~16GB). In practice, RoBERTa consistently outperforms BERT on downstream NLP tasks. The disadvantage is that RoBERTa is marginally heavier in terms of training time. For this project, RoBERTa is preferred due to its stronger generalization, which is directly relevant to the domain transfer objective.

Dataset: Jigsaw Toxic Comment Classification

The Jigsaw dataset contains approximately 160,000 comments from Wikipedia talk pages, labeled across six toxicity categories. A binary label is derived: a comment is toxic if any category is positive. A balanced subset of 3,000 samples was extracted to avoid class imbalance issues during fine-tuning.

Alternative: OLID (Offensive Language Identification Dataset). Used in SemEval-2019, OLID provides a hierarchical label system (offensive vs. non-offensive, targeted vs. untargeted). Its advantage is a more fine-grained taxonomy of abuse.

Its disadvantage is that it is Twitter-based, which would reduce the contrast between source and target domains needed for the domain transfer experiments. Jigsaw was chosen precisely because it comes from a different platform (news comment sections), making the domain gap more meaningful.

Framework: PyTorch and Hugging Face Transformers

PyTorch provides the tensor computation and backpropagation engine. Hugging Face Transformers provides the pre-trained RoBERTa model and tokenizer. The AdamW optimizer with a learning rate of 2e-5 is used, which is the standard setting for fine-tuning transformer models.

Alternative: TensorFlow / Keras. TensorFlow offers equivalent functionality and also integrates with Hugging Face. PyTorch was preferred for its more flexible and Pythonic API, its dominant adoption in NLP research, and its better native support for custom training loops, which gives more control over the fine-tuning process.

3. SSL-Related Focus

The Self-Supervised Learning component of this project is RoBERTa's pre-training via Masked Language Modeling (MLM). In MLM, 15% of input tokens are randomly masked and the model must predict the original tokens using the surrounding context. No human-labeled data is required: the supervision signal comes from the text itself, which is what makes it self-supervised.

This pre-training phase gives RoBERTa its general language understanding before any fine-tuning. When fine-tuned on the Jigsaw dataset, the model already understands grammar, semantics, and word relationships. The fine-tuning step only adapts the top classification layer to the specific task.

The connection to domain transfer is direct. A model pre-trained on general web text (SSL) and fine-tuned on one domain may fail on a different domain because the linguistic patterns, slang, and norms differ. This degradation across domains is the central research question of this project.

4. Milestone 2 – Implementation and Results

All three pipeline scripts were implemented and executed. The model was fine-tuned for one epoch on the training subset of 2,100 samples, running entirely on CPU (Intel Iris Xe, no NVIDIA GPU available). The reduced dataset size and single epoch were chosen to produce results within the hardware constraints of this stage.

Results after one epoch of fine-tuning (validation set, 300 samples):

- Training Loss: 0.3166
- Validation Accuracy: 0.9433
- Validation F1-Score: 0.9420

Test set results (600 samples, held-out):

- Test Accuracy: 0.9067
- Test F1-Score: 0.9082
- Clean class — Precision: 0.92 | Recall: 0.89 | F1: 0.91
- Toxic class — Precision: 0.89 | Recall: 0.92 | F1: 0.91

The high accuracy and F1 on a balanced validation set indicate that RoBERTa's pre-trained representations transfer effectively to toxicity classification even with minimal fine-tuning. These results served as the baseline for the domain transfer experiments carried out in Milestone 3.

5. Milestone 3 – Final Version

Milestone 3 builds on Milestone 2 by adding a second dataset from a different domain and training three separate models to measure how well each one transfers across domains.

5.1 Changes with respect to Milestone 2

Main changes from Milestone 2:

Dataset: Jigsaw only (3,000 samples) → Jigsaw full (159,571) + OLID (11,916)

Preprocessing: one generic pipeline → two separate pipelines (one per domain)

Class balancing: undersampling → class weights

Models trained: 1 → 3 (Solo Jigsaw / Solo OLID / Mixed)

Test sets: 1 → 3 (Jigsaw / OLID / Combined)

Execution: CPU local → Google Colab T4 GPU

Scripts: 3 (updated for dual-domain pipeline)

5.2 Updated Software Architecture

The pipeline has three standalone scripts, keeping the flat structure from Milestone 2. All scripts run on Google Colab with a T4 GPU.

- **code/preprocess.py**: loads both datasets, applies the two cleaning pipelines, computes class weights, saves train/val/test splits and the mixed training set.
- **code/train.py**: trains three RoBERTa-base models (Solo Jigsaw, Solo OLID, Mixed) with class weights and AdamW optimizer.
- **code/evaluate.py**: loads the three models and evaluates each one on the three test sets, printing the domain comparison table.

5.3 Datasets

Jigsaw Toxic Comment Classification: 159,571 Wikipedia comments with a binary toxic/clean label. Splits: 127,613 train / 15,934 validation / 15,950 test. Only 10.2% are toxic, so class weights are needed. 159,571 Wikipedia talk page comments. Binary label: toxic if any of the six toxicity columns is positive. No subsampling.

tweet_eval offensive (OLID): 11,916 tweets from Twitter labeled as offensive (OFF→1) or not (NOT→0). Test set: 860 samples. 33.1% are offensive. Downloaded from cardiffnlp/tweet_eval on HuggingFace.

Vocabulary overlap: 73.5% of OLID tokens also appear in the Jigsaw vocabulary. That means 1 in 4 Twitter tokens was never seen during Jigsaw training, which explains why the model struggles when crossing domains. The Jigsaw training vocabulary contains 263,520 unique tokens. The OLID vocabulary contains 25,561 tokens.

Mixed training set: Jigsaw train + OLID train combined → 139,529 samples.

5.4 Preprocessing

Two separate cleaning pipelines were used instead of a single one:

Jigsaw pipeline: lowercase, remove URLs, remove HTML tags, remove special characters, normalise whitespace.

OLID/Twitter pipeline: remove URLs, replace @mentions with USER, split hashtags (#HateSpeech → hate speech), lowercase, remove special characters. This keeps the meaning of Twitter-specific tokens instead of deleting them.

Class weights: computed with sklearn compute_class_weight for each dataset. For Jigsaw: weight 0 = 0.5566, weight 1 = 4.9165. Applied via CrossEntropyLoss so the model pays more attention to toxic/offensive examples.

5.5 Experimental Design

Three models were trained and each was tested on three different test sets:

Training conditions:

- Solo Jigsaw: trained on 127,613 Wikipedia samples
- Solo OLID: trained on 11,916 Twitter samples
- Mixed: trained on both combined (139,529 samples)

Test sets:

- Test A: Jigsaw test (15,950 samples)
- Test B: OLID test (860 samples)
- Test C: Combined (16,810 samples)

This lets us measure in-domain performance, cross-domain degradation, and whether training on mixed data helps.

Training: RoBERTa-base, 3 epochs, batch size 16, max length 128, lr 2e-5, AdamW, linear warmup, gradient clipping. Run on Google Colab T4 GPU.

5.6 Results

Domain comparison table (F1 score):

Trained on	Jigsaw test A (F1)	OLID test B (F1)	Combined C (F1)
Solo Jigsaw	0.8352	0.6932	0.8174
Solo OLID	0.6718	0.7087	0.6757
Mixed	0.8093	0.6909	0.7926

The Solo Jigsaw model scores F1 = 0.8352 on Jigsaw but drops to 0.6932 on OLID — a 14-point drop from switching domains. The Solo OLID model scores 0.7087 on OLID but only

0.6718 on Jigsaw. The Mixed model is the most stable across both: 0.8093 on Jigsaw, 0.6909 on OLID, and 0.7926 combined. Every single-domain model gets worse on the other domain. Mixed training avoids that at a small cost in peak performance.

5.7 Conclusions

Milestone 3 completes the domain transfer study started in Milestone 2. The key contributions with respect to Milestone 2 are:

- Used the full Jigsaw dataset (159,571 samples) and added OLID (11,916 tweets) instead of the small balanced subset from Milestone 2.
- Built two separate cleaning pipelines, one for Wikipedia text and one for Twitter text.
- Ran a 3×3 comparison: three models, three test sets, measuring in-domain and cross-domain performance.
- Replaced undersampling with class weights so the full dataset could be used.
- 73.5% vocabulary overlap confirms the domain gap: 1 in 4 Twitter tokens is unknown to the Jigsaw model.

The 28-point F1 drop confirms what Rosa et al. (2019) found: domain mismatch is the main reason these models fail. Waseem et al. (2017) also show that implicit abuse is the hardest to transfer across platforms, which matches what we see here.

References

- Rosa, H. et al. (2019). Automatic cyberbullying detection: A systematic review. Computers in Human Behavior, 93, 333–345.
- Waseem, Z. et al. (2017). Understanding abuse: A typology of abusive language detection subtasks. Proceedings of the First Workshop on Abusive Language Online.